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Ph.D. Candidate ◊ Department of Electrical and Computer Engineering ◊ CMU ◊ wadmes.github.io/cv/

HIGHLIGHTS

Vision: Pioneering **Autonomous EDA** via multi-modal agents and differentiable optimization to enable self-evolving design cycles with orders-of-magnitude acceleration.

Funding: Awarded **\$500,000+** in fellowships: Croucher (\$200k), Apple (\$100k×2), and Qualcomm (\$100k).

Impact: **4 Best Papers/Mentions** (ASP-DAC, ICLAD, etc.); NVIDIA Patent (Global Routing) ; Algorithms integrated into Apple production; Ongoing collaborations with Google, Intel, Broadcom, and more.

Leadership: Evaluated faculty candidates on the CMU ECE Faculty Search Committee; mentored 8+ students with placements at Stanford PhD, CMU PhD, and NVIDIA.

EDUCATION

Carnegie Mellon University, PA, USA	2021 – Present
Doctor of Philosophy , Department of Electrical and Computer Engineering Advisor: Professor Shawn Blanton & Professor José Moura	
The Chinese University of Hong Kong, Hong Kong	
Master of Philosophy , Department of Computer Science and Engineering Thesis: <i>Irregular Deep Data Embedding and Learning</i> (Faculty Outstanding Thesis Award) Supervisor: Professor Bei Yu	2019 – 2021
Bachelor of Science, ELITE Stream , Department of Computer Science and Engineering	2014 – 2018

SELECTED AWARDS AND HONORS

Fellowships & Scholarships

Croucher Fellowship	Croucher Foundation	2026
<i>Approx. \$200,000; 2-year stipend and research grant support. The most prestigious award for PhDs from HK.</i>		
Qualcomm Innovation Fellowship	Qualcomm Inc.	2024
<i>\$100,000 research grant; awarded for high-impact innovation globally in EDA.</i>		
Apple PhD fellowship in Integrated Systems	Apple Inc.	2024
<i>Approx. \$100,000; for elite IC PhD at designated premier universities; includes full funding and Apple mentorship.</i>		
Apple PhD fellowship in Integrated Systems	Apple Inc.	2022
<i>Approx. \$100,000; for elite IC PhD at designated premier universities; includes full funding and Apple mentorship.</i>		
Jack and Mildred Bowers Scholarship	CMU	2025
Dean's Fellowship	CMU	2021
Talent Development Scholarship	HKSAR Government	2021
Full Postgraduate Studentship	CUHK	2019-2021
ELITE Stream Student Scholarship	Faculty of Engineering, CUHK	2018
Undergraduate Admission Scholarship	Soong Ching Ling Foundation	2015-2018
<i>¥200,000; Top 0.1% in Provincial National College Entrance Examination (Gaokao)</i>		

Awards & Honors

Best Paper Award, Honorable Mention	ICLAD, top 3 out of 94 submissions	2025
<i>Invited for internal sharing at Apple (100+ attendees).</i>		
Best Paper Award	ASP-DAC, top 2 out of 368 submissions	2021
Best Student Paper Award	ICTAI, top 1 out of 458 submissions	2019
Distinguished Paper Award	ISSTA, top 3 out of 142 submissions	2019
Faculty Outstanding Thesis Award	Engineering Faculty, CUHK	2021
<i>Sole recipient of the best thesis award per year in the Faculty of Engineering..</i>		
1st Place Award in EDA elite challenge	Chinese Institute of Electronics	2020
Richard Newton Young Student Fellow	DAC	2020
2nd Place Award in CAD Contest	ICCAD	2018

PATENTS

- [P2] **Wei Li**, Rongjian Liang, Haoxing Ren. “Differentiable Global Router”. U.S. Patent Application 18/965,415, Filed Dec 2, 2024. Assigned to NVIDIA Corp. - Patent Pending.
- [P1] **Wei Li**, Ronald Blanton, Jose M. F. Moura. “System and Method for Global Floorplanning via Semidefinite Programming”. U.S. Patent Application 18/910,938, Filed: Oct 9, 2024 —Patent Pending.

PUBLICATIONS

Journal Papers

- [J2] **Wei Li**, Yuzhe Ma, Yibo Lin, Bei Yu, “Adaptive Layout Decomposition with Graph Embedding Neural Networks”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**).
- [J1] **Wei Li**, Yuzhe Ma, Qi Sun, Zhang Lu, Yibo Lin, Iris Hui-Ru Jiang, Bei Yu, David Z. Pan, “OpenMPL: An Open Source Layout Decomposer”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**).

Conference Papers

- [18] **Wei Li**, Yang Zou, Yixin Liang, Shawn Blanton and José Moura, “DEFT: Differentiable Framework for Circuit Testing”, ACM/IEEE Design Automation Conference (DAC), 2026, Under Review.
- [C17] **Wei Li**, Yang Zou, Christopher Ellis, Ruben Purdy, Shawn Blanton and José Moura, “BRIDGE: Bridging Graph and Large Language Models in EDA”, IEEE International Conference on LLM-Aided Design (**ICLAD**), 2025. (**Best Paper Award, Honorable Mention**)
- [C16] Chris Nigh, Ruben Purdy, **Wei Li**, Subhasish Mitra, R.D. Blanton, “ IC-PEPR: PEPR Testing Goes Intra-Cell”, IEEE International Test Conference (**ITC**), 2025.
- [C15] Ruben Purdy, Chris Nigh, **Wei Li**, R.D. Blanton, “CHEF: CHaracterizing Elusive Logic Circuit Failure” , IEEE VLSI Test Symposium (**VTS**) 2025.
- [C14] Chris Nigh, Ruben Purdy, **Wei Li**, Subhasish Mitra, R.D. Blanton, “Faulty Function Extraction for Defective Circuits” , IEEE European Test Symposium (**ETS**) 2024.
- [C13] **Wei Li**, Rongjian Liang, Anthony Agnesina, Haoyu Yang, Chia-Tung Ho, Anand Rajaram, Haoxing Ren, “DGR: Differentiable Global Routing”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, 2024.
- [C12] **Wei Li**, Ruben Purdy, Jose Moura, Shawn Blanton, “Characterize the ability of GNNs in attacking logic locking”, ACM/IEEE Workshop on Machine Learning for CAD (**MLCAD**), Snowbird, Utah, Sep. 11–13, 2023.
- [C11] **Wei Li**, Fangzhou Wang, Jose Moura, Shawn Blanton, “Global floorplanning via semidefinite programming”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, July 9-13, 2023.
- [C10] **Wei Li**, Chris Nigh, Danielle Duvalsaint, Subhasish Mitra, R.D. Blanton, “PEPR: Pseudo-Exhaustive Physical Region Testing”, IEEE International Test Conference (**ITC**), Sep. 25 - Sep. 30, 2022.
- [C9] **Wei Li**, Guojin Chen, Haoyu Yang, Ran Chen, Bei Yu, “Learning Point Clouds in EDA”, ACM International Symposium on Physical Design (**ISPD**), Mar. 21–Mar. 24, 2021.
- [C8] **Wei Li**, Yuxiao Qu, Gengjie Chen, Yuzhe Ma, Bei Yu, “TreeNet: Deep Point Cloud Embedding for Routing Tree Construction”, IEEE/ACM Asian and South Pacific Design Automation Conference (**ASP-DAC**), Tokyo, Jan. 18–21, 2021. (**Best Paper Award**)
- [C7] **Wei Li**, Jialu Xia, Yuzhe Ma, Jialu Li, Yibo Lin, Bei Yu, “Adaptive Layout Decomposition with Graph Embedding Neural Networks”, ACM/IEEE Design Automation Conference (**DAC**), San Francisco, July 19–23, 2020.
- [C6] Husheng Zhou, **Wei Li**, Yuankun Zhu, Yuqun Zhang, Bei Yu, Lingming Zhang, Cong Liu, “DeepBillboard: Systematic Physical-World Testing of Autonomous Driving Systems”, ACM/IEEE International Conference on Software Engineering (**ICSE**), Seoul, May 23–29, 2020.
- [C5] Yuzhe Ma, Zhuolun He, **Wei Li**, Tinghuan Chen, Lu Zhang, Bei Yu, “Understanding Graphs in EDA: From Shallow to Deep Learning”, ACM International Symposium on Physical Design (**ISPD**), Taipei, Mar. 25–Apr. 01, 2020.

- [C4] Yuzhe Ma, Ran Chen, **Wei Li**, Fanhua Shang, Wenjian Yu, Minsik Cho, Bei Yu, “A Unified Approximation Framework for Deep Neural Networks”, The IEEE International Conference on Tools with Artificial Intelligence (**ICTAI**) 2019. (**Best Student Paper Award**)
- [C3] **Wei Li**, Yuzhe Ma, Qi Sun, Yibo Lin, Iris Hui-Ru Jiang, Bei Yu, David Z. Pan, “OpenMPL: An Open Source Layout Decomposer”, IEEE International Conference on ASIC (**ASICON**), Chongqing, China, Oct. 29–Nov. 1, 2019.
- [C2] Xia Li, **Wei Li**, Yuqun Zhang, Yuqun Zhang, Lingming Zhang, “DeepFL: Integrating Multiple Fault Diagnosis Dimensions for Deep Fault Localization”, The ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA**), 2019. (**Distinguished Paper Award**)
- [C1] Bentian Jiang, Xiaopeng Zhang, Ran Chen, Gengjie Chen, Peishan Tu, **Wei Li**, Evangeline F. Y. Young, Bei Yu, “FIT: Fill Insertion Considering Timing”, ACM/IEEE Design Automation Conference (**DAC**), Las Vegas, NV, June 2-6, 2019.

RESEARCH EXPERIENCE - IN UNIVERSITY

PhD candidate, Carnegie Mellon University, United States

Aug. 2021 – Now

Multi-modal agentic AI for EDA

- Introduces a new paradigm where EDA agents move beyond rigid, predefined APIs and interact with a unified, multi-modal representation of the design environment.
- Outlines two challenging case studies—automated RTL coding and Timing ECO—to demonstrate the framework’s effectiveness.
- Investigates the new problem-solving strategies that emerge from this increased agent autonomy.

DEFT: Differentiable Automatic Test Pattern Generation

- Reformulated ATPG as a differentiable optimization with a new reparameterization
- Custom CUDA kernel achieves 4×–26× speedup on industrial circuits
- Improved HTD detection by 21%–49% under the same pattern budget
- Support practical ATPG features such as partial assignment, generating patterns with 19.3% fewer 0/1 bits while detecting 35% more faults

Graph Modality in LLMs for VLSI [ICLAD’25, Best Paper Honorable Mention]

- Pioneered the integration of graph modality into LLMs specifically for VLSI design.
- Engineered a fully automated, high-throughput data collection pipeline, curating a massive-scale dataset with over 10 billion training tokens.
- Garnered exceptional interest from industry, invited to deliver an internal technical talk at **Apple** to an audience of 100+ engineers.

Global Floorplanning using Semidefinite Programming [DAC’23]

- A SDP-based method for finding the best locations of modules in a chip; reduced wirelength by 3.02%–20.01%
- The industrial case study shows 500% quality improvement compared to the industrial tool.

Pseudo-Exhaustive Physically-Aware Region Testing [ITC’22]

- Comprehensively analyze both the physical layout and the logic netlist to identify single- or multi-output sub-circuits.
- Implemented a novel tensor-based representation of layout polygon coordinates that enables a neighborhood search strategy that reduces computational complexity from $O(n^2)$ to $O(dn)$.
- Implemented a GPU-based algorithm the physical sub-circuit extraction containing billions of sub-circuits.

GNN study in logic locking [MLCAD’23]

- Modeled their ability to identify circuit changes that stem from a logic lock as the ability to decide the isomorphism between logic netlists.
- Showed that GNNs are always upper bounded by heterogeneous Weisfeiler Lehman test in deciding the netlist isomorphism, and gave the conditions when GNNs reach the bound.

MPhil Student, The Chinese University of Hong Kong, Hong Kong

Aug. 2019 – May. 2021

Routing Tree Construction [ASP-DAC’21, Best Paper Award]

- Formalized special properties of the point cloud for the routing tree construction with theoretical proof.
- Proposed an adaptive flow, which used the cloud embedding obtained by a specifically-designed model based on special properties, to select the best approach and predict the best parameter.
- Outperformed previous methods by a large margin yet being extensible and flexible.

Adaptive Layout Decomposition [DAC’20, TCAD’21]

- Proposed an adaptive workflow for efficient decomposer selection and graph matching using graph embeddings.
- Designed a graph library construction algorithm based on graph embeddings for small graphs excluding isomorphic ones.
- Reduced the runtime by 87.7% while still preserving the optimality compared with the ILP-based decomposer.

Research Assistant, The Chinese University of Hong Kong, Hong Kong

Feb. 2019 – July. 2019

Open-source Layout Decomposition Framework [TCAD’21]

- Developed **OpenMPL**, a comprehensive open-source framework featuring high-performance implementations of state-of-the-art layout decomposition algorithms.
- Achieved **80+ GitHub stars**; the tool has been widely adopted as a baseline for benchmarking layout decomposition research.
- Identified critical algorithmic shortcomings in legacy heuristics and proposed optimized solutions that significantly improved decomposition success rates and scalability.

Acceleration and Compression of DNNs [ICTAI’19, Best Student Paper Award]

- Proposed a unified framework to compress CNNs by combining both lowrankness and sparsity.

- Compressed the model with up to $4.9\times$ reduction of parameters at a cost of little loss.

Research Assistant, Southern University of Science and Technology, China June. 2018 – Jan. 2019
Testing of Auto-driving Systems [ICSE'20]

- Introduced a joint optimization method to systematically generate adversarial perturbations to mislead steering of an autonomous driving system physically.
- Demonstrated the possibility of continuous physical-world tests for auto-driving scenarios as the first study.

Fault Localization [ISSTA'19, Distinguished Paper Award]

- Proposed a hierarchical DL approach to learn the most effective features for precise fault localization.
- Significantly outperformed state-of-the-art with over 20% improvement.

RESEARCH EXPERIENCE - IN INDUSTRY

Intern, SoC Physical Design Group, Apple, United States May. 2024 – Aug. 2024
Floorplan Encoder

- Propose a novel floorplan encoder for the floorplanning task, the encoder is capable of encoding the multi-modal and multi-objects floorplan state.
- Achieves 95% accuracy and shows 3X speedup to achieve the same quality compared to the industrial tool.

Research Intern, Nvidia, United States May. 2023 – Aug. 2023
Differentiable Global Routing [DAC'24]

- A differentiable global router capable of concurrent optimization for millions of nets
- Reduced nets with overflow by more than 80%
- Resulted in a patent filing with NVIDIA Research; officially released as an [open-source framework](#) via NVIDIA Labs.

Intern, SoC Physical Design Group, Apple, United States June. 2022 – Sep. 2022
Exploration of GNNs for Physical Design

- Implemented a basic GNN model for predicting holder buffer before routing.
- Evaluated multiple architectures, including path-based, sub-circuit based, and sub-graph based models.

Perfect Rectilinear Floorplanning

- A Simulated Annealing based algorithm for perfect rectilinear floorplanning.
- Reinforcement learning, and supervised-learning that guides SA are also explored.
- Integrated into physical design flows for large-scale SoC development.

TEACHING EXPERIENCE

Carnegie Mellon University
Head Teaching Assistant

18-765: Digital System Testing and Testable Design Fall 2025

- Curated and coordinated a guest seminar series featuring **10+ industry leaders** from **Apple, Broadcom, Qualcomm, Siemens, and Synopsys** to bridge theoretical testing concepts with industrial practices.
- Architected and implemented an **automated grading system**, ensuring consistent feedback for students.
- Led weekly office hours and technical Q&A; coordinated the grading team to synchronize curriculum delivery.

18-202: Mathematical Foundations of Electrical Engineering Spring 2023

- Conducted weekly recitations to simplify complex mathematical concepts for students.
- Managed course logistics, grading, and student consultations.

Teaching Assistant

18-765: Digital System Testing and Testable Design Fall 2022, Fall 2023, Fall 2024

- Developed and refined course assignments; provided in-depth technical support for projects.

The Chinese University of Hong Kong

Head Teaching Assistant

CENG2030: Fundamentals of Embedded Systems Spring 2021

CENG3420: Computer Organization and Design Spring 2020

- Designed and supervised weekly laboratory tutorials, focusing on hardware-software co-design and system-level performance.

MENTORING EXPERIENCE

Undergraduate Students Mentored

Aug 2024 - Present	Yang Zou [C17, 18]	CMU B.S.	Stanford Ph.D. soon
Aug 2019 - May 2020	Yuxiao Qu [C8]	CUHK B.S.	CMU Ph.D.
Aug 2019 - Dec 2019	Jialu Li [C7]	CUHK B.S.	HKUST M.Phil.
Aug 2019 - Dec 2019	Jialu Xia [C7]	CUHK B.S.	Spring Labs.

Graduate Students Mentored

Sep 2025 - Present	Rebecca Dettmar	CMU M.S.	In progress
Sep 2025 - Present	Zifeng Wang	CMU M.S.	In progress
Sep 2025 - Present	Andrew Kim	CMU M.S.	In progress
Sep 2025 - Jan 2026	Yixin Liang [18]	CMU M.S.	NVIDIA

PROFESSIONAL SERVICE

Journal & Conference Reviewer

- **Journals:** IEEE TCAD, IEEE TIFS, ACM TODAES, The ANZIAM Journal, Integration (the VLSI Journal).
- **Conferences** (Sub-reviewer/External): DAC, ITC, VTS, ETS.

Departmental Service

Student Representative (Volunteer), CMU ECE Faculty Search Committee 2025

- Participated in candidate interviews and social evaluations for Tenure-Track Assistant Professor applicants;
- Provided student-perspective feedback to the hiring committee.

Volunteer, CMU ECE PhD Open House 2025

- Mentored prospective PhD students and introduced research work;

PhD Student Contact 2026

- Provided insights to admitted PhD students about research and life at CMU;
- facilitated their transition into the program.

SELECTED INVITED TALKS & PANELS

2026.03	Towards Autonomous EDA: Research and Vision	Apple
2026.03	Introduction to Graduate Studies - Research Panel	CMU, ECE Department
2026.02	Differentiable ATPG w. Mathematical Foundations	Apple
2026.01	Towards Autonomous EDA: Research and Vision	Ricursive Intelligence (Founding Team)
2025.08	When LLMs Meet Graphs	Apple
2025.01	Review on ATPG and Fault Modeling	NVIDIA Research
2024.11	Differentiable Global Routing	Apple